

Anshuman Goel

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EDUCATION

Thapar Institute of Engineering and Technology
B.E. Computer Science, CGPA: 7.5

Patiala, India
Aug 2023 - May 2027 (Expected)

Guru Nanak Foundation Public School
Senior Secondary (CBSE) Board, Class 12th

Patiala, India
Apr 2021 - Mar 2023

PROJECTS

Concord - Multi-Agent Research Platform | Python, FastAPI, LangGraph, React, PostgreSQL, Docker

GitHub

- Built **Concord**, an **AI-powered multi-agent research platform** where specialised AI agents collaboratively retrieve, verify, and synthesise information into **citation-backed research reports**.
- Designed a distributed architecture using **FastAPI**, **LangGraph**, PostgreSQL (pgvector), Docker, and LLMs to orchestrate scalable multi-agent workflows.
- Optimised workflow execution through **intelligent retry logic** and **intent-aware follow-up handling**, reducing unnecessary LLM API calls by **65%**.

VerifyHub | React, JavaScript / TypeScript, Supabase, PostgreSQL

GitHub

- Built **VerifyHub**, a **full-stack certificate verification platform** that eliminates manual verification by automatically matching student identities across **Coursera certificates** and **LinkedIn profiles** in real time.
- Combined **parallel web automation**, a custom **Jaccard-based fuzzy identity matching algorithm**, and **Supabase Edge Functions** to accurately automate certificate validation through **real-time cross-platform identity matching**.
- Engineered secure verification workflows with **SSRF protection**, JWT replay prevention, and rate limiting, reducing manual verification effort by over **90%** across **12,000+** student submissions.

Distributed In-Memory Cache | C++, REST APIs, Multithreading, Distributed Systems

GitHub

- Built a **distributed in-memory caching system** that provides scalable, fault-tolerant, and low-latency data storage through intelligent request distribution across multiple cache nodes.
- Implemented **consistent hashing** with configurable **virtual nodes** and a modular **Strategy-based architecture**, reducing key distribution imbalance by **85%** (7% → 1%) across **10,000 keys** while scaling from **10** to **500 virtual nodes**.
- Engineered **thread-safe concurrent access** using reader-writer locks and atomic operations, sustaining over **682,000 operations/sec** under an **8-thread** workload, while exposing cache and cluster management through **RESTful APIs** with automatic node failure recovery and zero observed data races.

WORK EXPERIENCE

Software Engineering Intern | SafeSpace

May 2025 - July 2025

- Engineered **SafeSpace**, an **AI-powered mental health platform**, by developing a multimodal stress detection pipeline using CNNs, DNNs, and a novel **Agreement-Aware Fusion (AAF)** algorithm, while integrating **Whisper** and the **Phi-2 LLM** to deliver real-time personalized wellness recommendations, achieving **94.1%** accuracy on the RAVDESS dataset.

SKILLS

- Languages:** Java, Python, C++, JavaScript, TypeScript, SQL, HTML, CSS
- Web Development:** Tailwind CSS, React, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JWT Authentication
- Databases & Tools:** PostgreSQL, MySQL, MongoDB, Redis, Supabase, Docker, AWS, Git, GitHub, pgvector
- AI & ML:** LangGraph, TensorFlow, PyTorch, Whisper, Matplotlib, Plotly
- Core CS:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design (LLD), Agile Methodologies

ACHIEVEMENTS

- Innovation Award - Cyber AI Hackathon 2025** (University of Derby) - Recognized for building an AI-driven pharmacy intelligence platform enabling early epidemic detection and smart inventory management using real-world, imperfect data.
- Competitive Programming** - **400+** problems solved across LeetCode & other platforms; LeetCode Contest Rating **1,885** (Top 5.2%).

LEADERSHIP & EXTRACURRICULARS

Head of Logistics | Mudra Cultural Society & MUN Society, Thapar Institute

Aug 2023 – May 2026

- Led logistics and operations for large-scale cultural events and Model United Nations conferences, coordinating teams, vendors, and resources for events with over **1,200+** attendees and **300+** delegates.
- Optimized event planning, resource allocation, and registration workflows, improving operational efficiency while reducing logistics costs by **20%**.